20CYS433	BLOCKCHAIN TECHNOLOGY	L-T-P-C:3-0-0-3

Prerequisite(s): Nil

Course Objectives

- To provide an understanding of Decentralized blockchain-based systems, such as Bitcoin and Ethereum, and its position in the present technological landscape.
- To understand the impact and role of Blockchain Technology in financial, information, and other infrastructures. This course covers the technical aspects of public distributed ledgers, blockchain systems, cryptocurrencies, and smart contracts

Course Outcome

CO1: Understand the basic principles of Distributed Ledger Technology

CO2: Able to demonstrate the cryptographic primitives in Blockchain technology

CO3: Understand and Evaluate various consensus protocols

CO4: Develop Smart Contracts and create a DApp using Ethereum Blockchain

CO5: Analyze a real-world use case and provide how blockchain could be used to address the challenges faced

CO-PO Mapping

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	1	2									2	2
CO2	3	3	3	3	3								3	3
CO3	2	3	3	3									2	2
CO4	3	2	3	3	2								2	3
CO5	3	3	3	3	2							2	3	3

Syllabus

Blockchain Data Structure – Hash Chain - Distributed Database - Blockchain Architecture - Terminologies in Blockchain: Hashes - Transactions - Addresses - Wallet - Private Key Storage - Ledgers - Blocks - Chaining Blocks; Consensus and multiparty agreements: Proof of Work (PoW) - Proof of Stake (PoS) - Delegated Proof of Stake (DPoS) - Proof of Elapsed Time (PoET) - Proof of Importance - Reputation-based mechanisms - Practical Byzantine Fault Tolerance (PBFT); Blockchain Platforms: Cryptocurrencies (Bitcoin, Litecoin, Ethereum) -

Hyperledger - Ethereum; Blockchain implementation; Smart Contract - Web3.js - MetaMask; Forking; Soft Fork - Hard Fork - Cryptographic Changes and Forks; Blockchain as a Service - IPFS and Blockchain - Challenges in Blockchain; Concurrency, Scalability and Privacy.

Text Book(s)

1. *Imran Bashir, Mastering Blockchain; 2017.*
2. *Narayanan, J. Bonneau, E. Felten, A. Miller, S. Goldfeder, Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction, Princeton Univ Press; 2016*
3. *Alex Leverington, Ethereum Programming, Packt Publishing Limited; 2017.*

Reference(s)

1. *Andreas M. Antonopoulos, Mastering Bitcoin - Programming the Open Blockchain, O'Reilly Media, Inc.; 2017*
2. *Draft NISTIR 8202, Blockchain Technology Overview - NIST CSRC; 2018.*
3. *Roger Wattenhofer, CreateSpace, The Science of the Blockchain, Independent Publishing Platform; 2016*

Evaluation Pattern (50:50)

Assessment	Internal	External
Periodical 1	15	
Periodical 2	15	
*Continuous Assessment (CA)	20	
End Semester		50

*CA – Can be Quizzes, Assignments, Projects, and Reports